

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

TEAM ID	SWTID2026-3387
PROJECT TITLE	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
MAX MARKS	5 MARKS

4.3 – Solution Architecture

1. Data Sources

The solution starts with collecting housing market data from Excel files, CSV files, SQL databases, or publicly available housing datasets. These datasets contain information such as sale price, bedrooms, bathrooms, house area, basement area, year built, property condition, grade, and location.

2. Data Collection & Extraction

The collected data is connected to Tableau using its built-in data connectors. Tableau extracts the required records while validating the completeness and consistency of the dataset.

3. Data Preparation

Before visualization, the data is cleaned by removing duplicate records, handling missing values, correcting data types, creating calculated fields, and transforming the dataset into a structured format suitable for analysis.

4. Tableau Data Model

The cleaned dataset is organized into dimensions and measures. Relationships and hierarchies are defined where necessary to enable efficient analysis and drill-down operations.

5. Data Visualization Layer

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Tableau creates multiple visualizations such as:

- Bar Charts for price comparison.
- Line Charts for trend analysis.
- Pie Charts for category distribution.
- Scatter Plots for relationship analysis.
- Heat Maps for geographical or density insights.
- Histograms for price distribution.
- Tree Maps for hierarchical comparisons.

6. Interactive Dashboard

The individual worksheets are combined into an interactive dashboard. Users can apply filters, parameters, and dashboard actions to explore housing data dynamically and gain meaningful insights.

7. Tableau Storyboard

A Tableau Story is developed to present the complete analysis in a logical sequence. It guides users through the dataset overview, visual analysis, market trends, key observations, and final conclusions.

8. Web Integration

The completed Tableau dashboard is published and embedded into a Flask web application, allowing users to access interactive housing market analysis through any modern web browser.

9. Project Output

The final output includes an interactive Tableau dashboard, a Tableau Story, analytical reports, project documentation, and decision-support insights that help users understand housing market trends and property price behavior